Flutter Exam – 10 Questions

Question 1: Structure of a Flutter App

Explain the basic structure of a Flutter application.

Your answer should include:

- `main()` function
- `runApp()`
- `MaterialApp`
- `Scaffold`
- `build()` method

Task:

Write a simple Flutter program that displays:

```
Hello S2 Exam
```

in the center of the screen.

Question 2: Screen Representation

In Flutter, screens are typically represented using widgets.

Task:

Create a screen that contains:

- AppBar title: **Student Profile**
- Student Image
- Name
- Email
- Phone Number

Use appropriate widgets such as:

- `Column`
- `Text`
- `Image`
- `Padding`

Question 3: Custom Widget

Explain what a Custom Widget is in Flutter.

Task:

Create a reusable widget named:

```
ProfileItem
```

This widget must display:

Title + Value

Example:

```
Mastering Java : $25.00
Java Programming: $12.90
```

Then use the widget 3 times in a screen.

Question 4: Input Form

Flutter provides widgets to collect user input.

Task:

Create a Registration Form containing:

- Name
- Email
- Password
- Confirm Password
- Register Button

Requirements:

- Use `TextFormField`
- Use `Form`
- Validate empty fields

Question 5: State Management (Local State)

Explain the concept of Local State in Flutter.

Task:

Create a Counter App that:

- Displays a number
- Has a button **Increase**
- Each click increases the number by 1

Use:

`setState()`

Question 6: State Management using Provider

Flutter apps often use Provider for state management.

Task:

Create a simple Cart Counter app.

Requirements:

- Use `Provider`
- Display number of items in cart
- Button **Add to Cart**
- The number should update automatically

Explain the role of:

- `ChangeNotifier`
- `ChangeNotifierProvider`
- `Consumer`

Question 7: Login with REST API

Flutter apps commonly authenticate users through REST APIs.

Task:

Create a Login Screen containing:

- Email
- Password
- Login Button

When the user presses login, send a POST request to:

`https://example.com/api/login`

Request body:

```
{
  "email": "...",
  "password": "..."
}
```

Explain how to:

- Send request using `http`
- Decode JSON response
- Navigate to Home screen if success

Question 8: CRUD Operation using REST API

Flutter can interact with backend APIs to perform CRUD operations.

Task:

Create a Product List App that:

1. GET products from API
2. Display them in `ListView`
3. Add new product
4. Edit product
5. Delete product

Example API:

```
GET    /products
POST   /products
PUT    /products/{id}
DELETE /products/{id}
```

Explain which HTTP method is used for each operation.

Question 9: Firebase Firestore

Flutter can store data using Cloud Firestore.

Task:

Create a Student List App using Firestore.

Each student document contains:

- `name`
- `email`
- `phone`

Requirements:

- Add student
- Show student list
- Delete student

Explain how to use:

- `FirebaseFirestore.instance.collection()`
- `add()`
- `snapshots()`

Question 10: Local Storage and Navigation

Flutter apps often store local data and navigate between pages.

Task:

Create an app with:

Page 1: Login

- Username field
- Save username to `SharedPreferences`

Page 2: Home

- Display saved username
- Logout button

Requirements:

- Use `SharedPreferences`
- Use `Navigator.push()`
- Use `Navigator.pop()`

Explain how navigation works in Flutter.

Flutter Exam – Concise Answers (Each as a Full App)

Q1: Structure of a Flutter App

- **`main()`** – entry point
- **`runApp()`** – attaches root widget to screen
- **`MaterialApp`** – sets up Material Design & routing
- **`Scaffold`** – provides AppBar, body layout
- **`build()`** – returns the widget UI tree

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        body: Center(
          child: Text('Hello S2 Exam', style: TextStyle(fontSize: 24)),
        ),
      ),
    );
  }
}
```

Q2: Screen Representation

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

```

```
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) => MaterialApp(home: StudentProfileScreen());
}

class StudentProfileScreen extends StatelessWidget {
  const StudentProfileScreen({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Student Profile')),
      body: Padding(
        padding: EdgeInsets.all(16),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.center,
          children: [
            CircleAvatar(radius: 50, backgroundImage: NetworkImage('https://i.pravatar.cc/150')),
            SizedBox(height: 12),
            Text('John Doe', style: TextStyle(fontSize: 22, fontWeight: FontWeight.bold)),
            Text('john@example.com', style: TextStyle(color: Colors.grey)),
            Text('+855 12 345 678', style: TextStyle(color: Colors.grey)),
          ],
        ),
      ),
    );
  }
}

```

Q3: Custom Widget

A **Custom Widget** is a reusable class extending `StatelessWidget` or `StatefulWidget`, accepting data via constructor parameters.

```
import 'package:flutter/material.dart';
```

```
2
3 void main() => runApp(const MyApp());
4
5 class MyApp extends StatelessWidget {
6   const MyApp({super.key});
7   @override
8   Widget build(BuildContext context) => MaterialApp(home: ProductScreen());
9 }
10
11 // Custom Widget
12 class ProfileItem extends StatelessWidget {
13   final String title;
14   final String value;
15   const ProfileItem({super.key, required this.title, required this.value});
16
17   @override
18   Widget build(BuildContext context) {
19     return Padding(
20       padding: EdgeInsets.symmetric(vertical: 8),
21       child: Row(
22         mainAxisAlignment: MainAxisAlignment.spaceBetween,
23         children: [
24           Text(title, style: TextStyle(fontWeight: FontWeight.bold, fontSize:
16)),
25           Text(value, style: TextStyle(fontSize: 16)),
26         ],
27       ),
28     );
29   }
30 }
31
32 class ProductScreen extends StatelessWidget {
33   const ProductScreen({super.key});
34   @override
35   Widget build(BuildContext context) {
36     return Scaffold(
```

```
      appBar: AppBar(title: Text('Products')),
      body: Padding(
        padding: EdgeInsets.all(16),
        child: Column(
          children: [
            ProfileItem(title: 'Mastering Java', value: '\$25.00'),
            ProfileItem(title: 'Java Programming', value: '\$12.90'),
            ProfileItem(title: 'Flutter Basics', value: '\$19.99'),
          ],
        ),
      ),
    );
  }
}

```

Q4: Input Form

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) => MaterialApp(home: RegisterScreen());
}

class RegisterScreen extends StatefulWidget {
  const RegisterScreen({super.key});
  @override
  State<RegisterScreen> createState() => _RegisterScreenState();
}

class _RegisterScreenState extends State<RegisterScreen> {
  final _formKey = GlobalKey<FormState>();
  final _passwordController = TextEditingController();
```

```
20
21  @override
22  Widget build(BuildContext context) {
23    return Scaffold(
24      appBar: AppBar(title: Text('Register')),
25      body: Padding(
26        padding: EdgeInsets.all(16),
27        child: Form(
28          key: _formKey,
29          child: Column(
30            children: [
31              TextFormField(decoration: InputDecoration(labelText: 'Name'),
32                  validator: (v) => v!.isEmpty ? 'Required' : null),
33              TextFormField(decoration: InputDecoration(labelText: 'Email'),
34                  validator: (v) => v!.isEmpty ? 'Required' : null),
35              TextFormField(
36                controller: _passwordController,
37                decoration: InputDecoration(labelText: 'Password'),
38                obscureText: true,
39                validator: (v) => v!.isEmpty ? 'Required' : null,
40              ),
41              TextFormField(
42                decoration: InputDecoration(labelText: 'Confirm Password'),
43                obscureText: true,
44                validator: (v) {
45                  if (v!.isEmpty) return 'Required';
46                  if (v != _passwordController.text) return 'Passwords do not
match';
47                  return null;
48                },
49              ),
50              SizedBox(height: 20),
51              ElevatedButton(
52                onPressed: () {
53                  if (_formKey.currentState!.validate()) {
54                    ScaffoldMessenger.of(context).showSnackBar(
```

```
                      SnackBar(content: Text('Registered!')),
                    );
                  }
                },
                child: Text('Register'),
              ),
            ],
          ),
        ),
      ),
    );
  }
}

```

Q5: State Management (Local State)

Local State = state owned by one widget, updated with `setState()` which triggers `build()` to re-run.

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) => MaterialApp(home: CounterScreen());
}

class CounterScreen extends StatefulWidget {
  const CounterScreen({super.key});
  @override
  State<CounterScreen> createState() => _CounterScreenState();
}

class _CounterScreenState extends State<CounterScreen> {
  int _count = 0;
```

```

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Counter')),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text('$_count', style: TextStyle(fontSize: 64)),
            SizedBox(height: 20),
            ElevatedButton(
              onPressed: () => setState(() => _count++),
              child: Text('Increase'),
            ),
          ],
        ),
      ),
    );
  }
}

```

Q6: State Management using Provider

Class	Role
`ChangeNotifier`	Holds state; calls `notifyListeners()` on change
`ChangeNotifierProvider`	Injects the notifier into the widget tree
`Consumer`	Rebuilds UI when `notifyListeners()` is called

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

void main() => runApp(
  ChangeNotifierProvider(
    create: (_) => CartProvider(),
    child: const MyApp(),
  ),
```

```
 9 );
10
11 class CartProvider extends ChangeNotifier {
12   int _count = 0;
13   int get count => _count;
14   void addItem() { _count++; notifyListeners(); }
15 }
16
17 class MyApp extends StatelessWidget {
18   const MyApp({super.key});
19   @override
20   Widget build(BuildContext context) => MaterialApp(home: CartScreen());
21 }
22
23 class CartScreen extends StatelessWidget {
24   const CartScreen({super.key});
25   @override
26   Widget build(BuildContext context) {
27     return Scaffold(
28       appBar: AppBar(title: Text('Cart')),
29       body: Center(
30         child: Consumer<CartProvider>(
31           builder: (context, cart, _) => Column(
32             mainAxisAlignment: MainAxisAlignment.center,
33             children: [
34               Text('Items: ${cart.count}', style: TextStyle(fontSize: 28)),
35               SizedBox(height: 20),
36               ElevatedButton(
37                 onPressed: cart.addItem,
38                 child: Text('Add to Cart'),
39               ),
40             ],
41           ),
42         ),
43       ),
44     );
```

```
  }
}
```

pubspec.yaml: `provider: ^6.1.2`

Q7: Login with REST API

- **`http.post()`** – sends POST request with JSON body
- **`jsonDecode()`** – parses JSON response
- **`Navigator.pushReplacement()`** – navigates to Home on success

```dart
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) => MaterialApp(home: LoginScreen());
}

class LoginScreen extends StatefulWidget {
  const LoginScreen({super.key});
  @override
  State<LoginScreen> createState() => _LoginScreenState();
}

class _LoginScreenState extends State<LoginScreen> {
  final _emailController = TextEditingController();
  final _passwordController = TextEditingController();

  Future<void> _login() async {
    final res = await http.post(
      Uri.parse('https://example.com/api/login'),
      headers: {'Content-Type': 'application/json'},
      body: jsonEncode({
```

```
28          'email': _emailController.text,
29          'password': _passwordController.text,
30        }),
31      );
32      if (res.statusCode == 200) {
33        Navigator.pushReplacement(context, MaterialPageRoute(builder: (_) =>
HomeScreen()));
34      } else {
35        ScaffoldMessenger.of(context).showSnackBar(SnackBar(content: Text('Login
failed')));
36      }
37    }
38
39    @override
40    Widget build(BuildContext context) {
41      return Scaffold(
42        appBar: AppBar(title: Text('Login')),
43        body: Padding(
44          padding: EdgeInsets.all(16),
45          child: Column(
46            children: [
47              TextField(controller: _emailController, decoration:
InputDecoration(labelText: 'Email')),
48              TextField(controller: _passwordController, decoration:
InputDecoration(labelText: 'Password'), obscureText: true),
49              SizedBox(height: 20),
50              ElevatedButton(onPressed: _login, child: Text('Login')),
51            ],
52          ),
53        ),
54      );
55    }
56  }
57
58  class HomeScreen extends StatelessWidget {
59    const HomeScreen({super.key});
60    @override
```

```
  Widget build(BuildContext context) => Scaffold(
    appBar: AppBar(title: Text('Home')),
    body: Center(child: Text('Welcome!')),
  );
}
```

pubspec.yaml: `http: ^1.2.1`

Q8: CRUD with REST API

Operation	HTTP Method	Endpoint
Read	GET	/products
Create	POST	/products
Update	PUT	/products/{id}
Delete	DELETE	/products/{id}

```
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

const base = 'https://example.com/api';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) => MaterialApp(home: ProductListScreen());
}

class ProductListScreen extends StatefulWidget {
  const ProductListScreen({super.key});
  @override
  State<ProductListScreen> createState() => _ProductListScreenState();
}

class _ProductListScreenState extends State<ProductListScreen> {
  List _products = [];
```

```
23
24  @override
25  void initState() { super.initState(); _getProducts(); }
26
27  Future<void> _getProducts() async {
28    final res = await http.get(Uri.parse('$base/products'));
29    setState(() => _products = jsonDecode(res.body));
30  }
31
32  Future<void> _addProduct() async {
33    await http.post(Uri.parse('$base/products'),
34        headers: {'Content-Type': 'application/json'},
35        body: jsonEncode({'name': 'New Product', 'price': 9.99}));
36    _getProducts();
37  }
38
39  Future<void> _editProduct(int id) async {
40    await http.put(Uri.parse('$base/products/$id'),
41        headers: {'Content-Type': 'application/json'},
42        body: jsonEncode({'name': 'Updated', 'price': 19.99}));
43    _getProducts();
44  }
45
46  Future<void> _deleteProduct(int id) async {
47    await http.delete(Uri.parse('$base/products/$id'));
48    _getProducts();
49  }
50
51  @override
52  Widget build(BuildContext context) {
53    return Scaffold(
54      appBar: AppBar(title: Text('Products')),
55      floatingActionButton: FloatingActionButton(onPressed: _addProduct, child:
Icon(Icons.add)),
56      body: ListView.builder(
57        itemCount: _products.length,
```

```
        itemBuilder: (_, i) {
          final p = _products[i];
          return ListTile(
            title: Text(p['name']),
            subtitle: Text('\$${p['price']}'),
            trailing: Row(mainAxisSize: MainAxisSize.min, children: [
              IconButton(icon: Icon(Icons.edit, color: Colors.blue), onPressed: () => _editProduct(p['id'])),
              IconButton(icon: Icon(Icons.delete, color: Colors.red), onPressed: () => _deleteProduct(p['id'])),
            ]),
          );
        },
      ),
    );
  }
}
```

pubspec.yaml: `http: ^1.2.1`

Q9: Firebase Firestore

Method	Purpose
`.collection('students')`	Reference to Firestore collection
`.add(data)`	Add new document with auto ID
`.snapshots()`	Real-time stream; UI auto-updates

```
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'package:cloud_firestore/cloud_firestore.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});
```

```
  @override
  Widget build(BuildContext context) => MaterialApp(home: StudentListScreen());
}

class StudentListScreen extends StatefulWidget {
  const StudentListScreen({super.key});
  @override
  State<StudentListScreen> createState() => _StudentListScreenState();
}

class _StudentListScreenState extends State<StudentListScreen> {
  final _col = FirebaseFirestore.instance.collection('students');
  final _nameCtrl = TextEditingController();
  final _emailCtrl = TextEditingController();
  final _phoneCtrl = TextEditingController();

  Future<void> _add() async {
    await _col.add({'name': _nameCtrl.text, 'email': _emailCtrl.text, 'phone': _phoneCtrl.text});
    _nameCtrl.clear(); _emailCtrl.clear(); _phoneCtrl.clear();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Students')),
      body: Column(children: [
        Padding(
          padding: EdgeInsets.all(8),
          child: Column(children: [
            TextField(controller: _nameCtrl, decoration: InputDecoration(labelText: 'Name')),
            TextField(controller: _emailCtrl, decoration: InputDecoration(labelText: 'Email')),
            TextField(controller: _phoneCtrl, decoration: InputDecoration(labelText: 'Phone')),
            SizedBox(height: 8),
```

```
            ElevatedButton(onPressed: _add, child: Text('Add Student')),
          ]),
        ),
        Expanded(
          child: StreamBuilder<QuerySnapshot>(
            stream: _col.snapshots(),
            builder: (_, snap) {
              if (!snap.hasData) return CircularProgressIndicator();
              final docs = snap.data!.docs;
              return ListView.builder(
                itemCount: docs.length,
                itemBuilder: (_, i) {
                  final d = docs[i].data() as Map<String, dynamic>;
                  return ListTile(
                    title: Text(d['name']),
                    subtitle: Text('${d['email']} • ${d['phone']}'),
                    trailing: IconButton(
                      icon: Icon(Icons.delete, color: Colors.red),
                      onPressed: () => _col.doc(docs[i].id).delete(),
                    ),
                  );
                },
              );
            },
          ),
        ),
      ]),
    );
  }
}
```

pubspec.yaml: `firebase_core`, `cloud_firestore`

Q10: Local Storage and Navigation

Navigation uses a stack: `push()` goes forward, `pop()` goes back.

```
import 'package:flutter/material.dart';
```

```
2 import 'package:shared_preferences/shared_preferences.dart';
3
4 void main() => runApp(const MyApp());
5
6 class MyApp extends StatelessWidget {
7   const MyApp({super.key});
8   @override
9   Widget build(BuildContext context) => MaterialApp(home: LoginPage());
10 }
11
12 // Page 1: Login
13 class LoginPage extends StatefulWidget {
14   const LoginPage({super.key});
15   @override
16   State<LoginPage> createState() => _LoginPageState();
17 }
18
19 class _LoginPageState extends State<LoginPage> {
20   final _controller = TextEditingController();
21
22   Future<void> _login() async {
23     final prefs = await SharedPreferences.getInstance();
24     await prefs.setString('username', _controller.text);
25     Navigator.push(context, MaterialPageRoute(builder: (_) => HomePage()));
26   }
27
28   @override
29   Widget build(BuildContext context) {
30     return Scaffold(
31       appBar: AppBar(title: Text('Login')),
32       body: Padding(
33         padding: EdgeInsets.all(16),
34         child: Column(children: [
35           TextField(controller: _controller, decoration:
InputDecoration(labelText: 'Username')),
36           SizedBox(height: 16),
```

```
37          ElevatedButton(onPressed: _login, child: Text('Login')),
38        ]),
39      ),
40    );
41  }
42 }
43
44 // Page 2: Home
45 class HomePage extends StatefulWidget {
46   const HomePage({super.key});
47   @override
48   State<HomePage> createState() => _HomePageState();
49 }
50
51 class _HomePageState extends State<HomePage> {
52   String _username = '';
53
54   @override
55   void initState() {
56     super.initState();
57     _load();
58   }
59
60   Future<void> _load() async {
61     final prefs = await SharedPreferences.getInstance();
62     setState(() => _username = prefs.getString('username') ?? 'Guest');
63   }
64
65   Future<void> _logout() async {
66     final prefs = await SharedPreferences.getInstance();
67     await prefs.remove('username');
68     Navigator.pop(context);
69   }
70
71   @override
72   Widget build(BuildContext context) {
```

```
    return Scaffold(
      appBar: AppBar(title: Text('Home')),
      body: Center(
        child: Column(mainAxisAlignment: MainAxisAlignment.center, children: [
          Text('Welcome, $_username!', style: TextStyle(fontSize: 24)),
          SizedBox(height: 20),
          ElevatedButton(onPressed: _logout, child: Text('Logout')),
        ]),
      ),
    );
  }
}
```

pubspec.yaml: `shared_preferences: ^2.3.2`